

Morphic Studio AI Intelligence & Knowledge Architecture

Document 72 — Long-Term Memory, Context Management & Knowledge Persistence Specification

Version: 1.0

Status: Approved

1. Purpose

This document defines the Long-Term Memory, Context Management, and Knowledge Persistence Framework for Morphe Studio.

Its purpose is to enable AI agents to retain, retrieve, and apply relevant knowledge across sessions, projects, organizations, and creative workflows while maintaining transparency, governance, privacy, and user control.

Memory serves as the foundation for continuous contextual intelligence.

2. Vision

Memory should function like an experienced creative collaborator.

Instead of simply recalling previous conversations, the system should understand:

- Projects
- Stories
- Characters
- Worlds
- User preferences
- Team conventions
- Organizational knowledge
- Historical creative decisions

Memory evolves alongside the creative process.

3. Design Principles

Memory follows these principles:

- Context over conversation history.
 - Relevant retrieval over exhaustive recall.
 - User ownership of memory.
 - Transparent memory usage.
 - Privacy by design.
 - Explainable retrieval.
 - Efficient scalability.
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4. Memory Layers

The platform maintains multiple memory layers:

Temporary Working Memory

- Current request
- Active reasoning
- Immediate AI operations

Session Memory

- Current editing session
- Recent actions
- Temporary workspace state

Project Memory

- Story Bible
- Characters
- World knowledge
- Assets
- Creative decisions

Organization Memory

- Shared templates
- Policies
- Standards

- Approved workflows

Persistent User Memory

- Preferences
- Frequently used workflows
- Saved prompt styles
- Personal defaults

Archived Memory

- Completed projects
- Historical versions
- Retired assets
- Previous releases

Each layer has independent lifecycle rules.

5. Context Assembly Pipeline

Before executing an AI task, the system assembles context from:

- Active workspace
- Current project
- Story Bible
- Relevant characters
- World lore
- Timeline
- Previous AI outputs
- User instructions
- Organizational policies

Only relevant information is included.

6. Memory Retrieval

Retrieval considers:

- Semantic similarity
- Project relevance
- Recency
- User intent
- Canon priority
- Workflow context

- Confidence scores

Retrieved knowledge is ranked before use.

7. Context Compression

Large contexts should be optimized through:

- Summarization
- Canon extraction
- Entity linking
- Relationship preservation
- Importance scoring
- Hierarchical representation

Compression should preserve creative meaning.

8. Knowledge Persistence

Persistent knowledge includes:

- Canon rules
- Character evolution
- World history
- Plot milestones
- Style guides
- Organizational standards
- AI-approved references

Persistence survives sessions and device changes.

9. Shared Team Memory

Organizations may maintain shared memory containing:

- Project conventions
- Team guidelines
- Shared prompts
- Workflow patterns
- Approved terminology
- Production standards

Permissions determine visibility and edit rights.

10. Memory Transparency

Users may inspect:

- Retrieved memories
- Active context
- Knowledge sources
- Memory confidence
- Retrieval rationale
- Recently accessed memories

Transparency strengthens trust in AI decisions.

11. Memory Management

Users may:

- Add memory
- Edit memory
- Archive memory
- Delete memory
- Pin important knowledge
- Merge duplicate memories
- Restore archived information

Memory management remains intuitive.

12. Memory Lifecycle

Knowledge progresses through:

Creation

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Validation

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Active Use



Optimization



Archive



Retention Review



Deletion (where appropriate)

Lifecycle policies vary by memory type.

13. Privacy Boundaries

Memory respects:

- User ownership
- Workspace isolation
- Organization boundaries
- Permission inheritance
- Data residency policies
- Legal requirements

AI agents access only authorized memory.

14. Cross-Agent Memory Sharing

Agents exchange knowledge through shared memory rather than isolated duplication.

Shared memory includes:

- Canon references
- Story relationships
- Workflow state

- Approved assets
- Creative constraints

Synchronization prevents inconsistent understanding.

15. Monitoring & Auditing

Track:

- Retrieval accuracy
- Memory utilization
- Context size
- Retrieval latency
- Memory freshness
- User corrections
- AI confidence

Audit records support continuous refinement.

16. Integration

The Memory Framework integrates with:

- Multi-Agent Orchestrator
- Knowledge Graph
- Story Workspace
- Character Studio
- World Builder
- Story Bible
- Workflow Engine
- Asset Manager
- Publishing Center
- Dashboard
- Analytics
- Administration

Memory becomes the shared intelligence layer across the platform.

17. Acceptance Criteria

The Memory Framework is successful if:

- AI retrieves relevant knowledge consistently.
 - Memory remains transparent and explainable.
 - Users control persistent memory.
 - Context assembly remains efficient.
 - Cross-agent collaboration shares knowledge effectively.
 - Privacy boundaries are enforced.
 - Memory scales with long-term creative projects.
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Conclusion

The Morphic Studio Long-Term Memory, Context Management, and Knowledge Persistence Framework establishes the platform's enduring intelligence layer. Through multi-tier memory, intelligent context assembly, transparent retrieval, user-controlled persistence, privacy-aware governance, and cross-agent knowledge sharing, the framework enables AI collaborators to accumulate meaningful understanding over time while preserving creator trust, creative continuity, and organizational scalability.